📄 Final Report: Food Delivery Time Prediction

🔷 1. Introduction

The objective of this project is to predict food delivery time based on various influencing factors such as distance, traffic conditions, weather, and delivery-related attributes. The project also includes classification of deliveries into “Fast” or “Delayed” categories using machine learning techniques.

🔷 2. Dataset Description

The dataset used in this project contains information related to food delivery operations. Key features include:

- **Distance** – Distance between restaurant and customer
- **Weather Conditions** – Weather during delivery
- **Traffic Conditions** – Traffic level affecting delivery
- **Vehicle Type** – Type of delivery vehicle used
- **Order Cost** – Cost of the order
- **Delivery Time** – Target variable (time taken for delivery)

Additional features such as location coordinates and time-based attributes may also be included to enhance prediction accuracy.

🔷 3. Data Preprocessing

To ensure data quality and improve model performance, the following preprocessing steps were applied:

- **Handling Missing Values**:
  Missing numerical values were filled using mean imputation, while categorical values were filled using the mode.

- **Encoding Categorical Variables**:
  Categorical features such as weather, traffic, and vehicle type were converted into numerical form using label encoding.
- **Feature Scaling**:
  Numerical features were standardized using StandardScaler to bring all values to a similar scale.
- **Feature Engineering**:
  - Created new features such as **Rush Hour** to capture time-based patterns.
  - Distance calculation was performed using geographical coordinates where required.
- **Outlier Detection**:
  Outliers were identified using boxplots and analyzed to reduce their impact on the model.

🔷 4. Exploratory Data Analysis (EDA)

EDA was performed to understand patterns and relationships in the dataset:

- **Descriptive Statistics** were used to summarize key numerical features.
- **Correlation Analysis** revealed strong relationships between delivery time and features such as distance and traffic conditions.
- **Visualization Techniques** such as heatmaps and boxplots were used to identify trends and outliers.

🔷 5. Model Development

Two machine learning models were implemented:

🔹 5.1 Linear Regression (Prediction Model)

- Used to predict the actual delivery time.
- Features such as distance, traffic, and order-related variables were used as inputs.

Evaluation Metrics:

- Mean Squared Error (MSE)
- Mean Absolute Error (MAE)
- R-squared (R^2)

🔹 5.2 Logistic Regression (Classification Model)

- Used to classify deliveries as **Fast** or **Delayed**.
- A binary target variable was created based on average delivery time.

Evaluation Metrics:

- Accuracy
- Precision
- Recall
- F1-score
- Confusion Matrix
- ROC Curve

🔷 6. Model Evaluation and Comparison

- The **Linear Regression model** provided continuous predictions of delivery time and performed well in identifying key influencing factors.
- The **Logistic Regression model** effectively classified deliveries into fast and delayed categories.

Comparison:

Model	Purpose	Strength
Linear Regression	Time Prediction	Quantitative accuracy
Logistic Regression	Classification	Decision-making support

Both models complement each other:

- Linear Regression helps estimate **exact delivery time**

- Logistic Regression helps in **operational decisions**

🔷 7. Key Insights

From the analysis, several important insights were observed:

- **Distance is the most significant factor** affecting delivery time.
- **Traffic conditions greatly increase delays**, especially during peak hours.
- **Weather conditions** also impact delivery efficiency.
- Deliveries during **rush hours** are more likely to be delayed.

🔷 8. Actionable Recommendations

Based on the findings, the following improvements are suggested:

1. **Optimize Delivery Routes**
   Use shortest-path algorithms or GPS-based optimization to reduce travel time.
2. **Improve Staffing During Peak Hours**
   Increase the number of delivery personnel during rush hours to handle high demand.
3. **Traffic-Aware Scheduling**
   Assign deliveries based on real-time traffic conditions.
4. **Training Delivery Personnel**
   Improve efficiency through better training and route familiarity.
5. **Dynamic Time Prediction**
   Use real-time data (traffic + weather) to provide more accurate delivery estimates.

🔷 9. Conclusion

This project successfully demonstrates how machine learning can be applied to predict delivery times and improve operational efficiency in food delivery systems. By combining regression and classification models, businesses can both estimate delivery times and make informed decisions to reduce delays and enhance customer satisfaction.